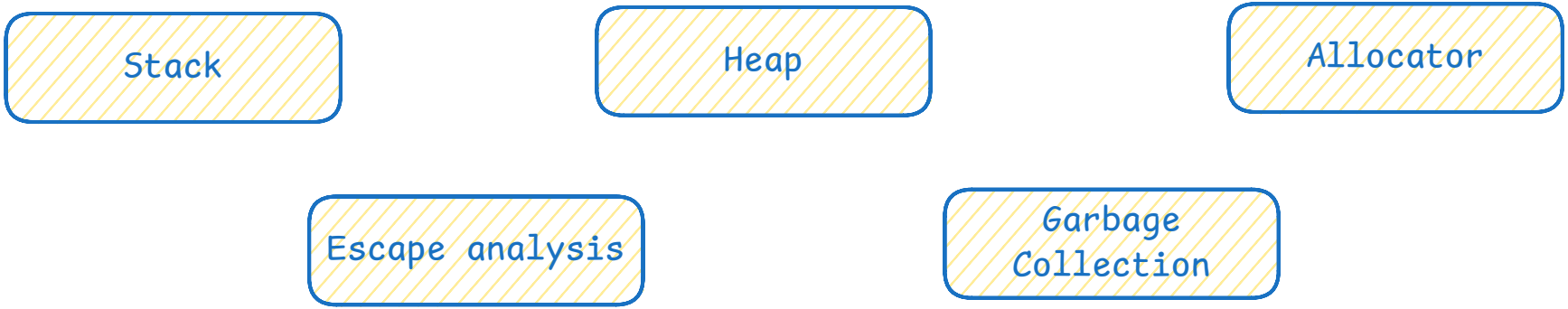
Lecture 07 - Memory Management

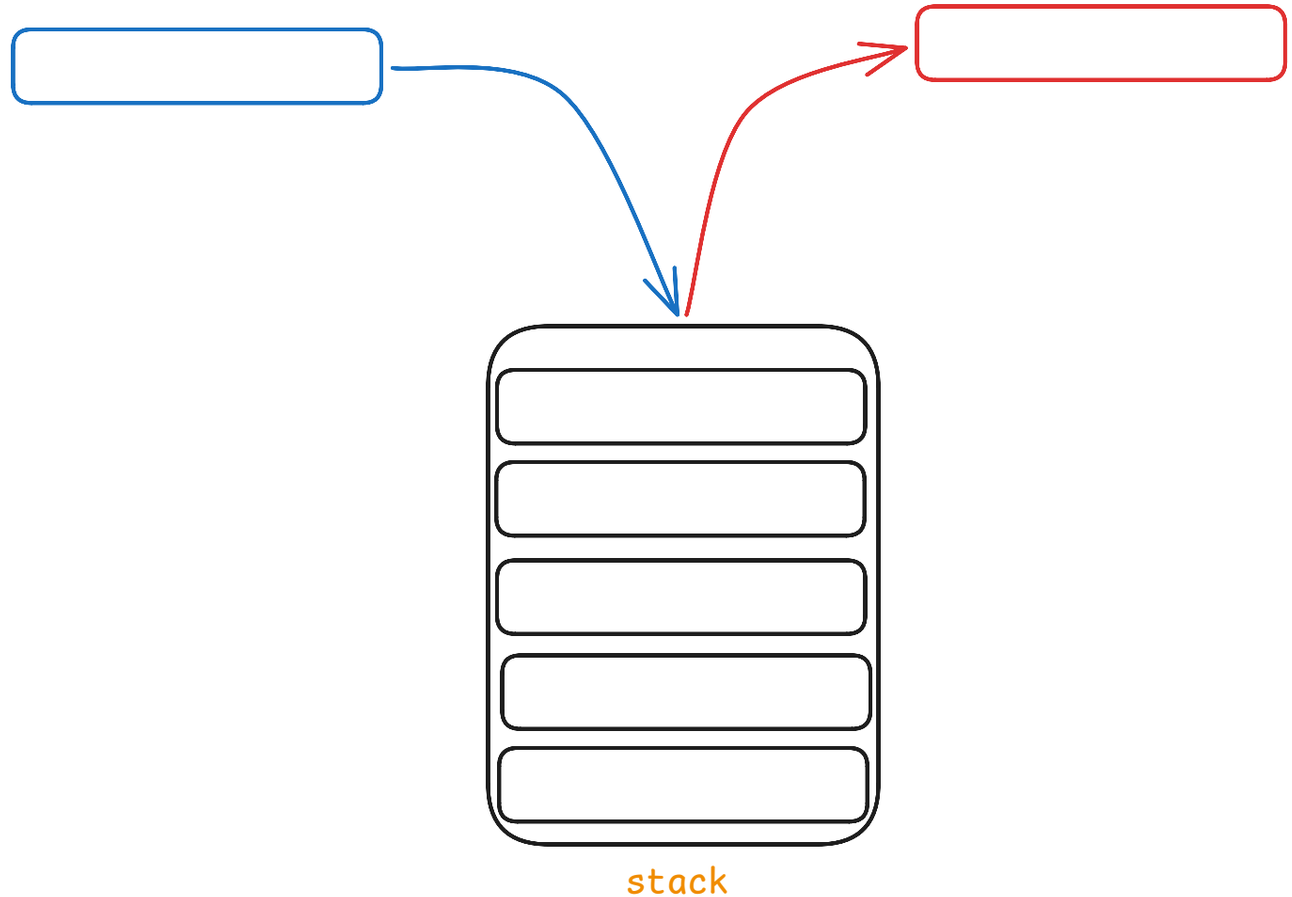


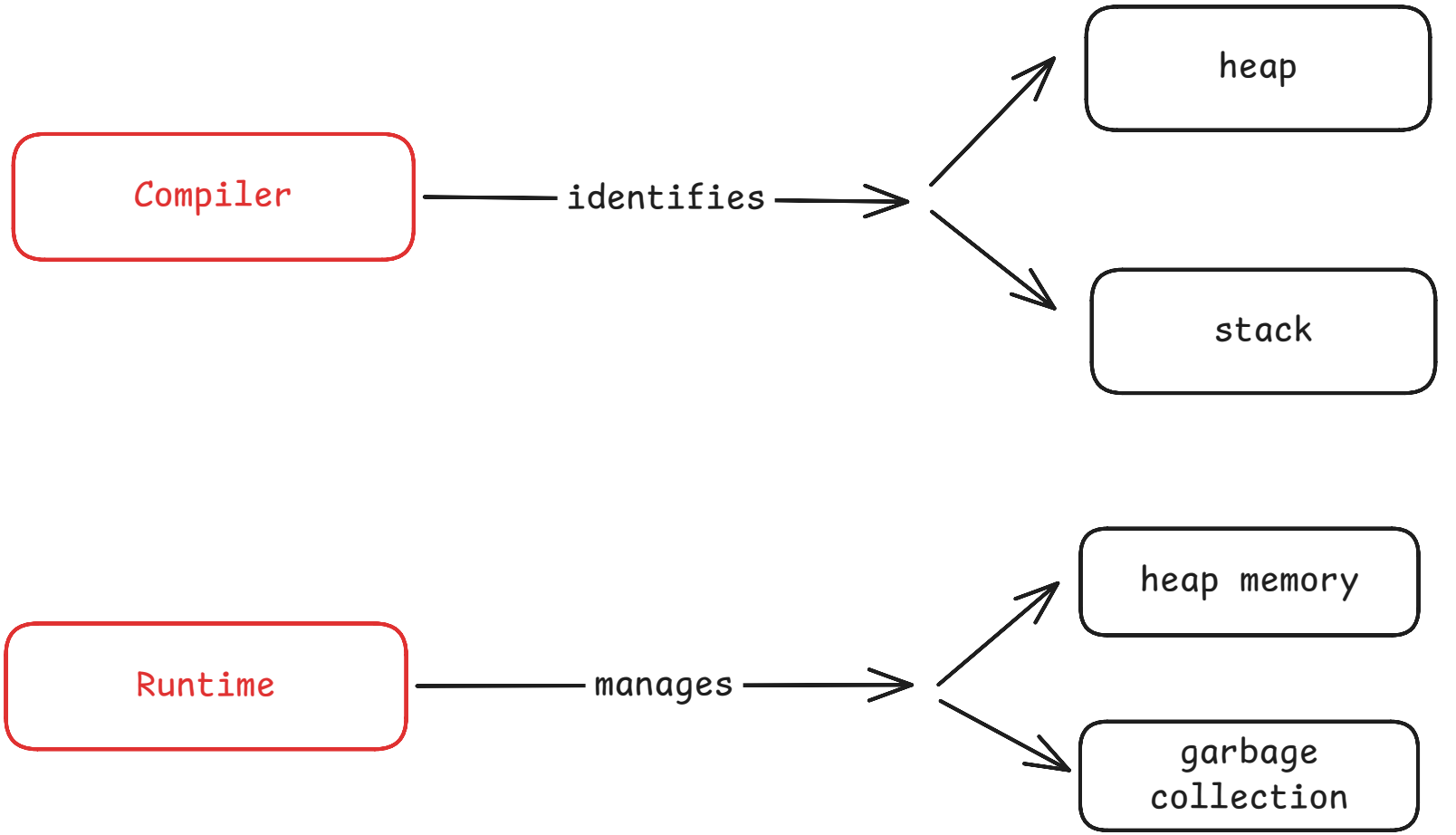
Memory Management

is the process of allocating, using, tracking, and reclaiming memory while a program is running.

Let's consider the following Go code:

```
func main() {
    age := 25
    name := "Alice"

    numbers := make([]int, 100)

    fmt.Println(age, name, numbers)
}
```

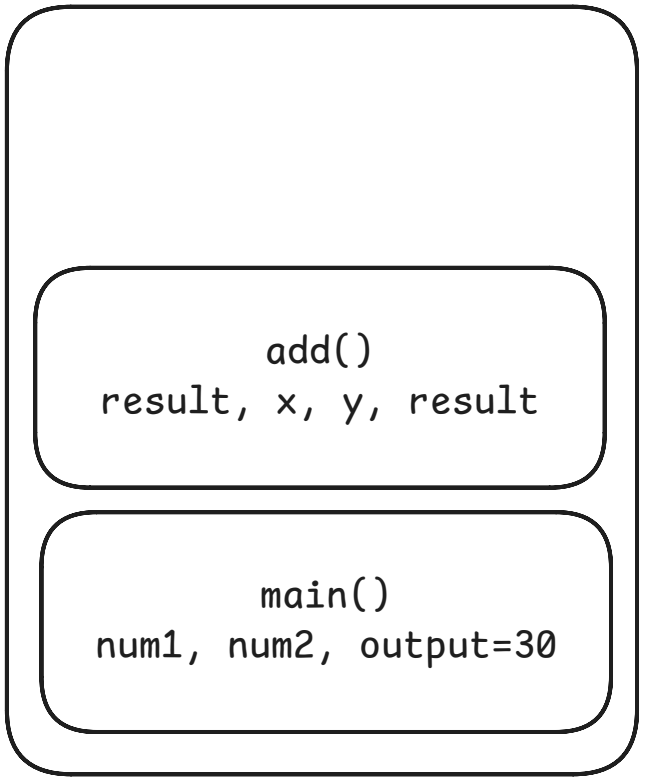



Compiler
identifies
heap
stack
Runtime
manages
heap memory
garbage collection

```
func add(x, y int) int {
    result := x + y
    return result
}

func main() {
    num1 := 10
    num2 := 20

    output := add(num1, num2)
    fmt.Println(output)
}
```

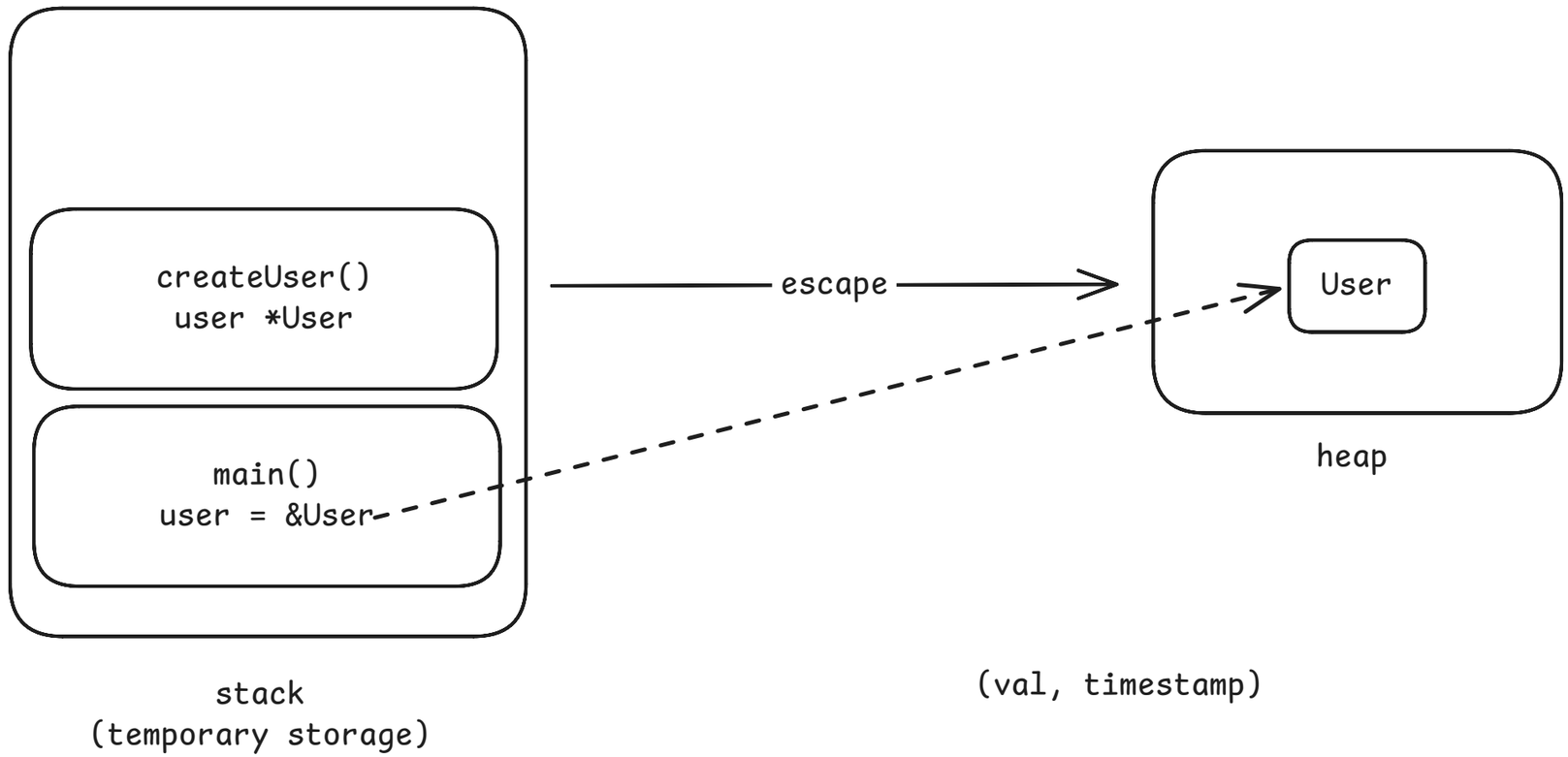
```
go run .
```

O(1)

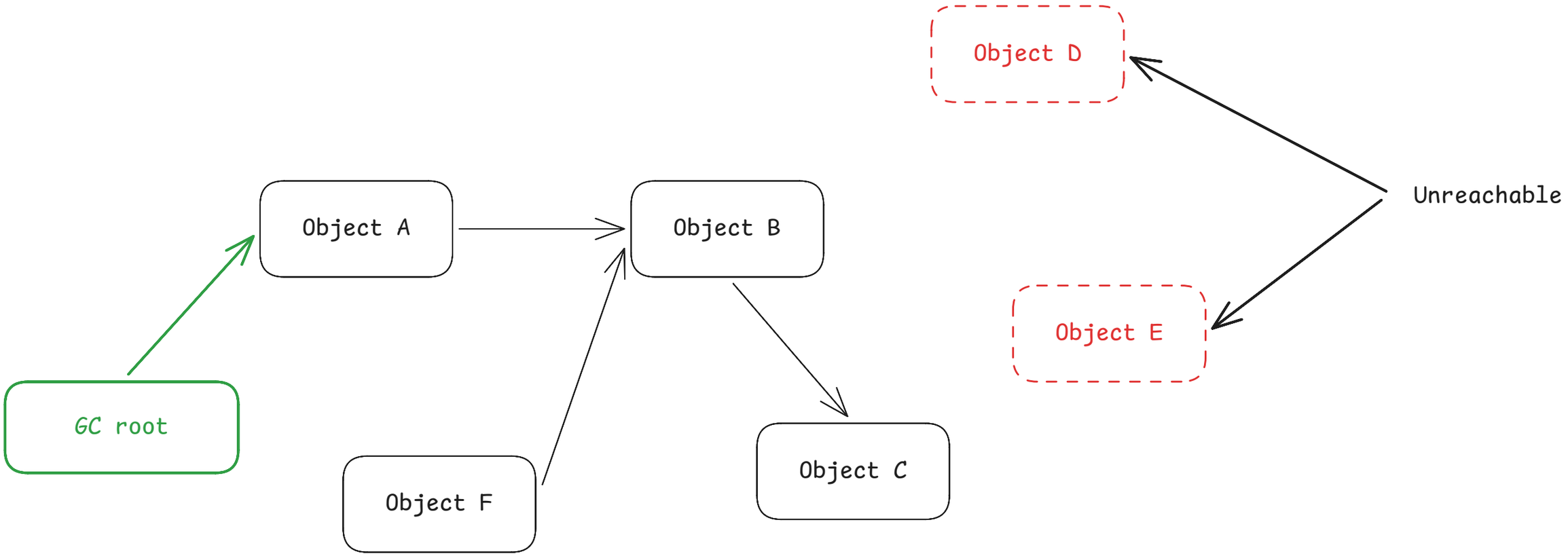


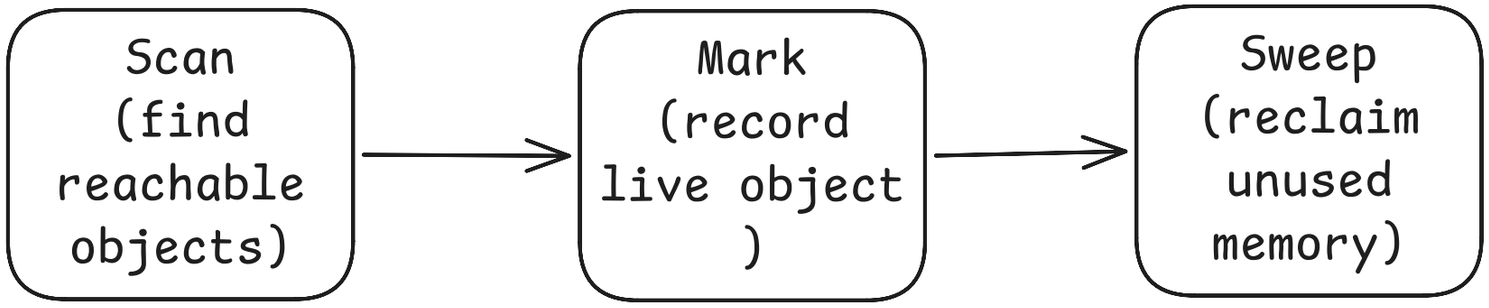
stack
(temporary storage)

```
type User struct {
    Name: string
    Age: int
}

func createUser() *User {
    user := User{
        Name: Nazar
        Age: 21
    }

    return &user
}

func main() {
    user := createUser()
}
```




```
func incremental(n *int) {
    *n = *n + 1
}

func main() {
    x := 10
    increment(&x)
    fmt.Println(x)
}
```


Object D
Unreachable
Object A
Object B
Object E
GC root
Object C
Object F


Scan (find reachable objects)
Mark (record live object)
Sweep (reclaim unused memory)